Digital-twin.isaac

Phase-1: Downloading and preparing requirements

1. Ubuntu
2. gazebo
3. isaac sim
4. ROS2

Phase-2: Building and animating a simple two-wheel mobile car in Isaac Sim using physics, rigid bodies, and revolute joints.

Step-1:

- After setting up requirements
- Open Ubundu Terminal and run:
  - `source /opt/ros/humble/setup.bash`
  - `/home/tiger/isaac-sim/isaac-sim.sh [path to your isaac sim]`
  - This will open isaac-sim that is connected with Ros2

Step-2:

Building simple mobile car (you can skip this part if you know to use isaac or you can build 3d models by code also !refer only if you want learn basic modeling)

1. set physics scene :- create→physics→physics scene
2. ground plane :- create→physics→ground plane
3. add cube shape for body :- create→shape→cube (name it as body)

4. You can adjust the size of the shapes on property panel on the right bottom part of the screen
5. add cylinder as wheel :- create→shape→cylinder (do this two times for two wheels, name it as left_wheel, right_wheel)
6. you can change the directions and size and make it look like wheel
7. select (body,left_wheel,right_wheel)→add+→physics→rigid with colliders preset
8. select (body,left_wheel)→right click on body→create→physics→joint→revolute joint [do same for right_wheel aswell]
9. select revolutejoint→local rotation 0 = 0, local rotation 1 = x -90 [same with revolute in right wheel]
10. change axis in both revolute to y

If you press play button, car should fall on the floor

12. select both revolutejoint→add+→physics→angular drive | drive→damping = 10000.0→target velocity = 100
13. create→material→omniPBR x2
14. rename to (body_material_color & wheel_material_color)
15. select body→none(near circle ball)→body_material_color [do this for wheels also]
16. select body_mate...→albedo→change to some color [same with wheel_material_color)

Simple mobile car with two wheels is now ready.

Phase-3: Build realistic mobile car according to the physical robot youre going to link with.

You can get codes for mobile cars in digital_twin_isaac folder.

1. copy the python code
2. go to window → script editor → paste the code in python container

Phase-4: Integrating the mobile car in Isaac Sim with ROS2 keyboard control using Action Graph and ROS2 bridge

Ensure that there is bridge connected between ubundu and isaac

- Open Isaac Sim. ,Go to **Window** → **Extensions**. →Search for `ROS 2`. →Make sure the **Omni.isaac.ros2_bridge** extension is toggled **ON** (enabled).

Step1

- Go to **Window** → **Visual Scripting** → **Action Graph**. A new window will appear at the bottom of your screen.
- Click the **New Action Graph** icon (a small page with a plus sign) in the middle of that window.
- On the left side of the Action Graph window, you will see a search bar. Search for and drag the following **four nodes** into the graph area:
  - `On Playback Tick`: This acts as the heartbeat, running the graph every frame.
  - `ROS2 Subscribe Twist`: This listens to the Ubuntu network for keyboard commands.
  - `Differential Controller`: This does the math to convert forward/turn commands into wheel speeds.
  - `Articulation Controller`: This physically forces your Revolute Joints to spin.

step2

- `On Playback Tick (Tick)` ➔ `ROS2 Subscribe Twist (Exec In)`
- `ROS2 Subscribe Twist (Exec Out)` ➔ `Differential Controller (Exec In)`

- `ROS2 Subscribe Twist (Linear Velocity)` ➔ `Break 3-Vector #1 (vector)` ➔ `(X)` ➔ `Differential Controller (linearVelocity)`
- `ROS2 Subscribe Twist (Angular Velocity)` ➔ `Break 3-Vector #2 (vector)` ➔ `(Z)` ➔ `Differential Controller (angularVelocity)`
- `On Playback Tick (Tick)` ➔ `Articulation Controller (Exec In)`
- `Differential Controller (Velocity Command)` ➔ `Articulation Controller (Velocity Command)`

step3

1. For `ROS2 Subscribe Twist`:

- **Node ID:** `ros2_twist` (or leave default).
- **Topic Name:** `/cmd_vel` *(This is the universal ROS 2 topic for moving robots).*

2. For `Differential Controller`:

- **Wheel Distance:** Enter the exact distance between your left and right wheels in meters (e.g., `0.15` for 15cm). *If this is wrong, your turns won't be accurate.*
- **Wheel Radius:** Enter the exact radius of your wheels in meters (e.g., `0.033` for 3.3cm).

3. For `Articulation Controller`:

- **Use Path:** Check this box (True).
- **Robot Path:** Click the target icon and select your robot's main folder (e.g., `/World/Robot_name`). *Note: This folder MUST have the Articulation Root API applied to it (Give Articulation Root API to `Robot_name` folder)*
- **Joint Names:** This is crucial. Type the exact names of your two revolute joints . It must look exactly like this: `your_left_wheel_joint_name , your_right_wheel_joint_name` In seperate boxes.

step4

- In Isaac Sim, press the **Play** button on the left toolbar. The car will drop onto the floor and wait.
- Open your **Ubuntu Terminal**.

- Source your ROS 2 installation by typing:
  `source /opt/ros/humble/setup.bash`
- Run the official ROS 2 keyboard controller by typing:
  `ros2 run teleop_twist_keyboard teleop_twist_keyboard`
- A menu will appear in your terminal showing you which keys to press (usually `i` to go forward, `j` and `l` to turn, `k` to stop).
- Keep your mouse clicked inside the Ubuntu Terminal, press `i`, and watch your digital twin drive forward on the screen.

Phase-5: Integrating with Joystick

Connecting with joystick. you dont have to change anything from the mapping of nodes to connect with joystick. it will auto calculate the calculations (only If keyboard node connections aleady present)

2nd Terminal

- `sudo apt update`
- sudo apt install ros-humble-joy ros-humble-teleop-twist-joy
- source /opt/ros/humble/setup.bash
- ros2 launch teleop_twist_joy teleop-launch.py

3rd Terminal

- source /opt/ros/humble/setup.bash
- ros2 topic echo /joy
- **Find the Deadman Switch:** Start pressing buttons on your SHANWAN controller one by one (R1, R2, Start, Select, X, Y, A, B).
- Watch the `buttons:` list in Terminal 2. When you see a `0` change to a `1` in the **9th spot** of that list (index 8), **THAT is your Deadman Switch.**
- **Drive the Car:** Now that you know which button it is, **press and hold** that specific button down.
- While holding it, push your **Left Joystick** straight up.

- Look at Isaac Sim—your car will drive forward